intervening space ensures the absence of a physical connection. Quantum mechanics challenges this view by revealing, at least in certain circumstances, a capacity to transcend space; long-range quantum connections can bypass spatial separation. Two objects can be far apart in space, but as far as quantum mechanics is concerned, it's as if they're a single entity. Moreover, because of the tight link between space and time found by Einstein, the quantum connections also have temporal tentacles. We'll shortly encounter some clever and truly wondrous experiments that have recently explored a number of the startling spatio-temporal interconnections entailed by quantum mechanics and, as we'll see, they forcefully challenge the classical, intuitive worldview many of us hold.

Despite these many impressive insights, there remains one very basic feature of time—that it seems to have a direction pointing from past to future—for which neither relativity nor quantum mechanics has provided an explanation. Instead, the only convincing progress has come from research in an area of physics called *cosmology.*

Cosmological Reality

To open our eyes to the true nature of the universe has always been one of physics' primary purposes. It's hard to imagine a more mind-stretching experience than learning, as we have over the last century, that the reality we experience is but a glimmer of the reality that is. But physics also has the equally important charge of explaining the elements of reality that we actually do experience. From our rapid march through the history of physics, it might seem as if this has already been achieved, as if ordinary experience is addressed by pre–twentieth-century advances in physics. To some extent, this is true. But even when it comes to the everyday, we are far from a full understanding. And among the features of common experience that have resisted complete explanation is one that taps into one of the deepest unresolved mysteries in modern physics —the mystery that the great British physicist Sir Arthur Eddington called the arrow of time.[4]

We take for granted that there is a direction to the way things unfold in time. Eggs break, but they don't unbreak; candles melt, but they don't